

Problem 1. Consider the equation

$$\lfloor x \rfloor (x^2 + 1) = x^3,$$

where $\lfloor x \rfloor$ denotes the greatest integer not exceeding x .

- (a) Prove that the equation has infinitely many positive real solutions.
- (b) Show that none of the positive real solutions of this equation is rational.

Solution. (a) Let $k = \lfloor x \rfloor$ and $y = x - k$. Then the equation becomes

$$k((k + y)^2 + 1) = (k + y)^3,$$

which reduces to

$$y(k + y)^2 = k.$$

The function $f(y) = y(k + y)^2$ is strictly increasing in $[0, 1]$ and continuous in the same interval. As $f(0) = 0 < k$ and $f(1) = (k + 1)^2 > k$, there exists exactly one $y_0 \in (0, 1)$ such that $f(y_0) = k$.

(b) The equation has no positive integral solutions. Assume that $x = k + y$ is rational and let $x = n/d$, where n and d are relatively prime positive integers. The given equation then becomes

$$\frac{k(n^2 + d^2)}{d^2} = \frac{n^3}{d^3},$$

or equivalently,

$$dk(n^2 + d^2) = n^3.$$

Since x is not an integer, d has at least one prime divisor. It follows from the last equation that this prime divisor also divides n , a contradiction. $\square$

Problem 2. Let n be a positive integer. Assume that n numbers are to be chosen from the table

0	1	$\cdots$	$n - 1$
n	$n + 1$	$\cdots$	$2n - 1$
$\vdots$	$\vdots$	$\ddots$	$\vdots$
$(n - 1)n$	$(n - 1)(n + 1)$	$\cdots$	$n^2 - 1$

with no two of them from the same row or the same column. Find the maximum value of the product of these n numbers.

Solution. The product is $R(\sigma) = \prod_{i=0}^{n-1} (ni + \sigma(i))$, for some permutation $\sigma: \{0, 1, \dots, n - 1\} \rightarrow \{0, 1, \dots, n - 1\}$. Let σ be such that $R(\sigma)$ is maximal. We may assume that all the factors $(ni + \sigma(i))$ are positive, because otherwise the product is zero, that is the smallest possible. Assume further that $\sigma(a) > \sigma(b)$ for some $a > b$. Let a permutation τ be defined by

$$\tau(i) = \begin{cases} \sigma(b), & i = a, \\ \sigma(a), & i = b, \\ \sigma(i), & \text{otherwise.} \end{cases}$$

Then we have

$$\frac{R(\tau)}{R(\sigma)} = \frac{(na + \tau(a))(nb + \tau(b))}{(na + \sigma(a))(nb + \sigma(b))} = \frac{(na + \sigma(b))(nb + \sigma(a))}{(na + \sigma(a))(nb + \sigma(b))} > 1$$

as

$$(na + \sigma(b))(nb + \sigma(a)) - (na + \sigma(a))(nb + \sigma(b)) = n(a\sigma(a) + b\sigma(b) - a\sigma(b) - b\sigma(a)) = n(a - b)(\sigma(a) - \sigma(b)) > 0.$$

This contradicts the maximality of $R(\sigma)$, hence σ has to satisfy $\sigma(a) < \sigma(b)$ for all $a > b$. Thus $\sigma(i) = n - 1 - i$ for all i , and so

$$R(\sigma) = \prod_{i=0}^{n-1} (ni + n - 1 - i) = \prod_{i=0}^{n-1} (i + 1)(n - 1) = (n - 1)^n n!.$$

$\square$

Problem 3. There are 2018 lamps arranged on a table. Two persons play the following game. In each move the player flips the switch of one lamp, but he must never get back an arrangement of the lit lamps that had happened before. A player who cannot move loses. Which player has a winning strategy?

Answer. The first player has a winning strategy.

Solution 1. The first player can pick one lamp and keep switching it on and off during the whole game. The second player cannot switch this particular lamp, he always has to switch some other lamp so that the arrangement of the other lamps becomes different from any that has already been on the table. So the first player always has a move, and the second player eventually runs out of the possible moves.

Solution 2. Note that the parity of the lit lamps changes with each move. So all the possible states can be divided into two disjoint sets, one with odd number of the lamps lit and the other with even number of the lamps lit. We get a bipartite graph where the vertices are the states and two states are connected with an edge if it is possible to get from one state to another by switching one lamp off or on.

We want to use Hall's marriage theorem to get a perfect matching of the states. The assumption of the theorem is the following: for every subset A of the states in one set there is at least as many neighbouring states in the second set. Let the number of states in A be n , and let B be the set of neighbouring states of A , containing m states. Since each state in A has exactly 2018 neighbours and all these neighbours belong to the set B , there are exactly $2018n$ edges between A and B . Since each state in B has exactly 2018 neighbours (some of them may not belong to A), there are at most $2018m$ edges between A and B . Hence $2018n \leq 2018m$, or $n \leq m$, i.e. the assumption of the theorem is satisfied.

By Hall's marriage theorem, there is a perfect matching. On every move the first player has to switch the lamp which changes the state into its partner in the perfect matching. Any lamp the second player can switch results in a state whose partner has not been used yet, so the first player always has a move, and the second player eventually loses. $\square$

Problem 4. Let T denote the 15-element set $\{10a + b : a, b \in \mathbb{Z}, 1 \leq a < b \leq 6\}$. Let S be a subset of T in which all six digits 1, 2, ..., 6 appear and in which no three elements together use all these six digits. Determine the largest possible size of S .

Answer. The largest possible size of S is 9.

Solution. Let $|S| = n$. Consider the numbers of T , which contain 1 or 2. Certainly, no 3 of them can contain all 6 digits and all 6 digits appear. Hence $n \geq 9$.

Now consider the partitions:

12,	36,	45,
13,	24,	56,
14,	26,	35,
15,	23,	46,
16,	25,	34.

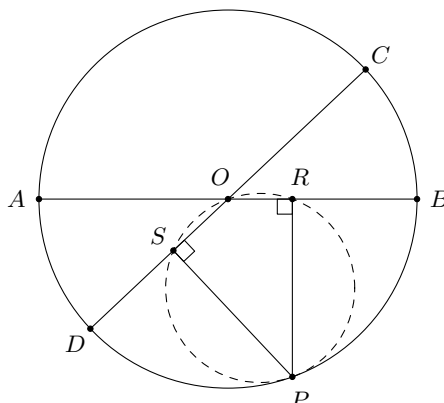
Since every row is a partition of $\{1, 2, \dots, 6\}$, it contains all 6 digits, S can contain at most two numbers of each of the 5 rows, i.e. $n \leq 10$.

Now we will prove that $n = 9$ is the correct number. Therefore we assume that $n = 10$ and will exclude this case by contradiction. Certainly, there is a digit, say 1, which does not appear at least twice (otherwise at most 3 numbers are missing in S) and at most 4 times (otherwise this digit does not appear in the members of S at all). Obviously, every row of the above set of partitions contains exactly 2 members of S . Without loss of generality, assume that $12, 13 \notin S$ and $16 \in S$. Then consider the following partitions, where boldfaced numbers are members of S and numbers in italics are not:

<i>12,</i>	36,	45,
<i>13,</i>	24,	56,
14,	26,	35,
15,	23,	46,
16,	25,	34.

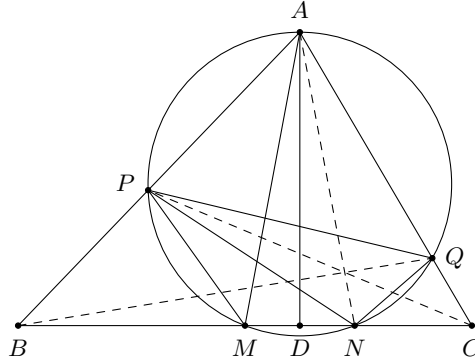
Since $16, 45 \in S$ it follows $23 \notin S$ and since $24, 36 \in S$ it follows $15 \notin S$. Now S is missing at least 2 members (15, 23) of the partition 15, 23, 46, which is a contradiction. $\square$

Problem 5. Let AB and CD be two diameters of the circle Γ . For an arbitrary point P on Γ , let R and S be the feet of the perpendiculars from P to AB and CD , respectively. Show that the length of RS is independent of the choice of P .



Solution. Let O be the centre of Γ . Then P, R, S , and O are points on a circle Γ' with diameter OP , which is equal to the radius of Γ . The segment RS is a chord of this circle subtending the angle BOD or a supplementary angle. Since the angle as well as radius of Γ' are independent of P , so is RS (just use the Extended Sine Formula). $\square$

Problem 6. In the non-isosceles triangle ABC an altitude from A meets side BC in D . Let M be the midpoint of BC and let N be the reflection of M in D . The circumcircle of triangle AMN intersects the side AB in $P \neq A$ and the side AC in $Q \neq A$. Prove that AN, BQ and CP are concurrent.



Solution 1. Without loss of generality, we assume the order of the points on BC to be B, M, D, N, C . This implies that P is on the segment AB and Q is on the segment AC . Since D is the midpoint of MN and AD is perpendicular to MN , the line AD is the perpendicular bisector of MN , which contains the circumcentre of $\triangle AMN$. As A is on the perpendicular bisector of MN , we have $AM = AN$. We now have $\angle APM = \angle AQN$. Therefore

$$\angle CQN = 180^\circ - \angle AQN = 180^\circ - \angle APM = \angle BPM$$

Furthermore, as $NMPQ$ is cyclic, we have

$$\angle NQP = 180^\circ - \angle NMP = \angle BMP$$

Hence

$$\angle AQP = 180^\circ - \angle CQN - \angle NQP = 180^\circ - \angle BPM - \angle BMP = \angle PBM = \angle ABC$$

Similarly,

$$\angle APQ = \angle BCA$$

Now we have $\triangle APQ \sim \triangle ACB$. So

$$\frac{AP}{AQ} = \frac{AC}{AB}$$

Furthermore, $\angle MAB = \angle MAP = \angle MNP = \angle BNP$, so $\triangle BMA \sim \triangle BPN$, and hence

$$\frac{BN}{BP} = \frac{BA}{BM}$$

We also have $\angle CAM = \angle QAM = 180^\circ - \angle QNM = \angle QNC$. This implies $\triangle CMA \sim \triangle CQN$, so

$$\frac{CQ}{CN} = \frac{CM}{CA}$$

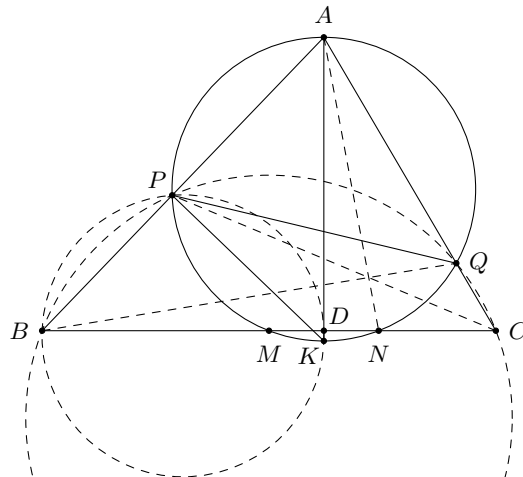
Putting everything together, we find

$$\frac{BN}{BP} \cdot \frac{CQ}{CN} \cdot \frac{AP}{AQ} = \frac{BA}{BM} \cdot \frac{CM}{CA} \cdot \frac{AC}{AB}$$

As $BM = CM$, the right-hand side is equal to 1. This means that

$$\frac{BN}{NC} \cdot \frac{CQ}{QA} \cdot \frac{AP}{PB} = 1.$$

By Ceva's theorem, AN, BQ and CP are concurrent. $\square$



Solution 2. We consider the same configuration as in Solution 1. Let K be the second intersection of AD with the circumcircle of $\triangle AMN$. Since D is the midpoint of MN and AD is perpendicular to MN , the line AD is the perpendicular bisector of MN , which contains the circumcentre of $\triangle AMN$. So AK is a diameter of this circumcircle. Now we have $\angle BPK = 90^\circ = \angle BDK$, so $BPDK$ is a cyclic quadrilateral. Also, A, M, N, K, P and Q are concyclic. Using both circles, we find

$$180^\circ - \angle CQP = \angle AQP = \angle AKP = \angle DKP = \angle DBP = \angle CBP$$

This implies that $BPQC$ is cyclic as well, and so we get $AP \cdot AB = AQ \cdot AC$, with directed lengths. Also, $BN \cdot BM = BP \cdot BA$ and $CN \cdot CM = CQ \cdot CA$. Hence

$$AP \cdot AB \cdot BN \cdot BM \cdot CQ \cdot CA = AQ \cdot AC \cdot BP \cdot BA \cdot CN \cdot CM$$

Changing the signs of all six lengths on the right-hand side and replacing MC by BM , we find

$$AP \cdot AB \cdot BN \cdot BM \cdot CQ \cdot CA = QA \cdot CA \cdot PB \cdot AB \cdot NC \cdot BM$$

Cleaning this up, we have

$$AP \cdot BN \cdot CQ = QA \cdot PB \cdot NC$$

implying

$$\frac{BN}{NC} \cdot \frac{CQ}{QA} \cdot \frac{AP}{PB} = 1.$$

By Ceva's theorem, AN, BQ and CP are concurrent. □

Problem 7. For any integer $n \geq 2$, we define A_n to be the number of positive integers m with the following property: the distance from n to the nearest multiple of m is equal to the distance from n^3 to the nearest multiple of m . Find all integers $n \geq 2$ for which A_n is odd. (Note: The distance between two integers a and b is defined as $|a - b|$.)

Solution. For an integer m we consider the distance d from n to the nearest multiple of m . Then $m \mid n \pm d$, which means $n \equiv \pm d \pmod{m}$. So if, for some m , the distance from n to the nearest multiple of m is equal to the distance from n^3 to the nearest multiple of m , then $n \equiv \pm n^3 \pmod{m}$. On the other hand, if $n \equiv \pm n^3 \pmod{m}$, then there exists a $0 \leq d \leq \frac{m}{2}$ such that $n \equiv \pm d \pmod{m}$ and $n^3 \equiv \pm d \pmod{m}$, so the distance from n to the nearest multiple of m is equal to the distance from n^3 to the nearest multiple of m .

We conclude that we need to count the number of positive integers m such that $n \equiv \pm n^3 \pmod{m}$, or, equivalently, $m \mid n^3 - n$ or $m \mid n^3 + n$. That is,

$$\begin{aligned} A_n &= |\{m \in \mathbb{Z}_{>0} : m \mid n^3 - n \text{ or } m \mid n^3 + n\}| \\ &= |\{m \in \mathbb{Z}_{>0} : m \mid n^3 - n\}| + |\{m \in \mathbb{Z}_{>0} : m \mid n^3 + n\}| - |\{m \in \mathbb{Z}_{>0} : m \mid n^3 - n \text{ and } m \mid n^3 + n\}| \\ &= |\{m \in \mathbb{Z}_{>0} : m \mid n^3 - n\}| + |\{m \in \mathbb{Z}_{>0} : m \mid n^3 + n\}| - |\{m \in \mathbb{Z}_{>0} : m \mid \gcd(n^3 - n, n^3 + n)\}| \\ &= \tau(n^3 - n) + \tau(n^3 + n) - \tau(\gcd(n^3 - n, n^3 + n)) \end{aligned}$$

where $\tau(k)$ denotes the number of (positive) divisors of a positive integer k .

Recall that $\tau(k)$ is odd if and only if k is a square. Furthermore, we have

$$\gcd(n, n^2 \pm 1) = 1.$$

So if $n^3 \pm n$ were a square, then both n and $n^2 \pm 1$ would be squares. But $n^2 \pm 1$ is not a square, since $n \geq 2$ and the only consecutive squares are 0, 1. Hence neither $n^3 - n$ nor $n^3 + n$ is a square, so the first two terms $\tau(n^3 - n)$ and $\tau(n^3 + n)$ are both even. Hence A_n is odd if and only if $\gcd(n^3 - n, n^3 + n)$ is a square.

We have

$$\gcd(n^2 - 1, n^2 + 1) = \gcd(n^2 - 1, 2) = \begin{cases} 1 & \text{if } n \text{ is even,} \\ 2 & \text{if } n \text{ is odd.} \end{cases}$$

Hence

$$\gcd(n^3 - n, n^3 + n) = \begin{cases} n & \text{if } n \text{ is even,} \\ 2n & \text{if } n \text{ is odd.} \end{cases}$$

Note that $2n$ for n odd is never a square, since it has exactly one factor of 2. We conclude that A_n is odd if and only if n is an even square. □

Problem 8. An integer $n \geq 1$ is called *balanced* if it has an even number of distinct prime divisors. Prove that there exist infinitely many positive integers n such that there are exactly two balanced numbers among $n, n + 1, n + 2$ and $n + 3$.

Solution. We argue by contradiction. Choose N so large that no $n \geq N$ obeys this property. Now we partition all integers $\geq N$ into maximal blocks of consecutive numbers which are either all balanced or not. We delete the first block from the following considerations, now starting from $N' > N$. Clearly, by assumption, there cannot meet two blocks with length ≥ 2 . It is also impossible that there meet two blocks of length 1 (remember that we deleted the first block). Thus all balanced or all unbalanced blocks have length 1. All other blocks have length 3, at least.

Case 1: All unbalanced blocks have length 1.

We take an unbalanced number $u > 2N' + 3$ with $u \equiv 1 \pmod{4}$ (for instance $u = p^2$ for an odd prime p). Since all balanced blocks have length ≥ 3 , $u - 3$, $u - 1$, and $u + 1$ must be balanced. This implies that $(u - 3)/2$ is unbalanced, $(u - 1)/2$ is balanced, and $(u + 1)/2$ is again unbalanced. Thus $\{(u - 1)/2\}$ is a balanced block of length 1, a contradiction.

Case 2: All balanced blocks have length 1.

Now we take a balanced number $b > 2N' + 3$ with $b \equiv 1 \pmod{4}$ (for instance $b = p^2 q^2$ for distinct odd primes p, q). By similar arguments, $(b - 3)/2$ is balanced, $(b - 1)/2$ is unbalanced, and $(b + 1)/2$ is again balanced. Now the balanced block $\{(b - 1)/2\}$ gives the desired contradiction. $\square$